Preety Jaiswal

M.Tech in AI for Sustainability
Indian Institute of Technology, Kanpur

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Education

Year	Degree/Certificate	Institute	CPI/%
2025–Present	M.Tech (AI for Sustainability)	Indian Institute of Technology, Kanpur	8.0/10
2023–2025	M.Sc (Statistics)	University of Allahabad, Prayagraj	8.9/10
2020–2023	B.Sc (Statistics)	University of Allahabad, Prayagraj	69.6%
2020	CBSE (Class XII)	St. Paul Public School, U.P.	87%
2018	UP Board (Class X)	Prestige Inter College, U.P.	88.2%

Research Experience

- **Physics-Guided Climate Modeling & Spatio-Temporal Analytics (M.Tech Thesis)** *Guide: Dr. Sarosh Ghausi(Ongoing)*
  - Evaluated a physics-constrained thermodynamic maximum power optimization model integrating Stefan-Boltzmann radiative transfer ($R_{L,\text{up}} = \epsilon\sigma T_s^4$) to dynamically solve turbulent heat fluxes and reconstruct surface temperature (T_s), achieving $R^2 > 0.93$ against ground-truth station observations.
  - Architected a high-throughput spatio-temporal data pipeline regridding and fusing 70+ years (1951–2021) of IMD climate reanalysis ($4{,}434$ spatial grid points) with 20+ years (2001–2020) of NASA CERES radiative-flux data at $1^\circ \times 1^\circ$ resolution, spanning 7,305 daily time steps.
  - Engineered an anomaly extraction pipeline using 11-day moving climatological baselines, OLS regression, and Mann-Kendall analysis to classify rainy vs.non-rainy regimes, quantifying warming divergence (**+0.2962°C/dec** vs. +0.1503°C/dec) and linking it to shortwave attenuation across **68.2%** of spatial grids.

Projects

- **ResearchMind: Enterprise Multi-Agent Deep Research System** (Self Project) *(May 2026)*
  - Architected a multi-agent deep research system with LangGraph/LangChain, orchestrating multi-query search, parallel source extraction, LLM synthesis, critic evaluation, and automatic self-correction for low-quality reports.
  - Engineered a high-throughput, fault-tolerant LLM pipeline using asyncio/aiohttp concurrent retrieval, TTL caching, and multi-model fallback routing across Gemini/OpenAI to reduce API calls and maintain reliable execution.
- **Pan-India Forest Fire Early-Warning & Risk Prediction System** (Course Project: KSS604) *(Feb – Apr 2026)*
  - Engineered a spatio-temporal wildfire risk prediction pipeline at 0.1° (∼11 km) resolution, fusing 13 years of ERA5, NDVI, and VIIRS observations with temporal lag, climatology, and environmental features for monthly fire-risk forecasting across India.
  - Benchmarked ConvLSTM2D, FT-Transformer, XGBoost, and Random Forest, achieving **0.958 ROC-AUC**, 0.899 PR-AUC, and 0.834 F_1 on 2022–2024 out-of-time data, with probability calibration, SHAP-based driver analysis.
- **Production AI Hybrid Movie Recommendation System** (Self Project) *(Jan 2026)*
  - Engineered a hybrid semantic recommendation engine using Sentence-Transformers to encode **55,000+** films into 384D dense embeddings, combining cosine similarity with Bayesian popularity scoring for personalized ranking.
  - Built an end-to-end ML pipeline with Precision@K, NDCG@K, and MRR benchmarking against a CountVectorizer baseline, integrating review sentiment and aspect analysis with Streamlit serving.
- **Cyclone Track Prediction** (Course Project: EE708) *(Sept – Nov 2025)*
  - Built a multi-horizon cyclone forecasting system (+6h to +72h) using 45+ years of NOAA IBTrACS data, engineering 149 meteorological, kinematic, temporal, and geospatial features with Coriolis dynamics, lag/rolling statistics, and displacement targets.
  - Benchmarked 7 ML models using a leakage-free chronological split (1980–2019 / 2020–2025), achieving **11.09 km RMSE** at 6h and **26.30 km** at 12h, reducing error by **40.9% and 39.7%** over Ridge.

Technical Skills

- **Programming Languages:** Python, SQL, R
- **Machine Learning & DL:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM, CNN, RNN, LSTM, Transformers
- **Generative AI & Agents:** LangChain, LangGraph, AI Agents, RAG, Sentence-Transformers, LLM Fine-Tuning
- **Data Science/ Libraries:**NumPy, Pandas, Matplotlib, Seaborn, GeoPandas, Feature Engineering, Time-Series Modeling
- **Tools & Platforms:** Streamlit, Docker, GitHub, Linux, LaTeX, Jupyter Notebook, Google Colab, VS code

Key Coursework

- **AI/ML & Data Mining:** Fundamentals of Data Science and Machine Learning, Knowledge Discovery & Data Mining, Regression Analysis, Statistical Computing (Python/R)
- **Statistical & Mathematical Foundations:** Multivariate Analysis, Measure & Probability Theory, Stochastic Processes, Estimation Theory, Hypothesis Testing